demonstrated transformation in Mice using *Diplococcus pneumoniae*. Two strains of this bacterium are present. One strain produces smooth colonies and are virulent in nature (S-type). In addition another strain produce rough colonies and are avirulent (R-type). When S-type of cells were injected into the mouse, the mouse died. When R-type of cells were injected, the mouse survived. He injected heat killed S-type cells into the mouse. The mouse did not die. When the mixture of heat killed S-type cells and R-type cells were injected into the mouse, the mouse died. The avirulent rough strain of *Diplococcus* had been transformed into S-type cells. The hereditary material of heat killed S-type cells had transformed R-type cell into virulent smooth strains. Thus the phenomenon of changing the character of one strain by transferring the DNA of another strain into the former is called Transformation.

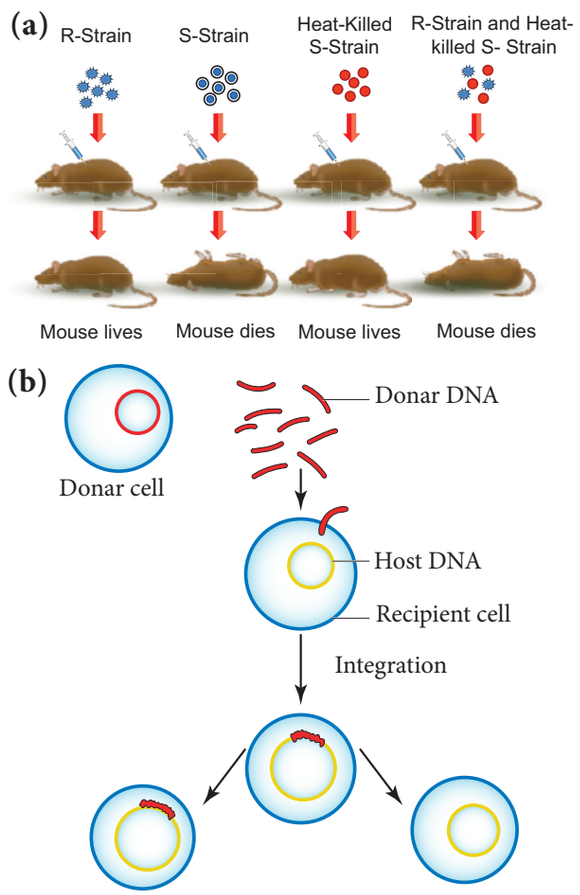


Figure 1.13: Transformation in Bacteria
(a) Griffith's experiment on Transformation
(b) Mechanism of Transformation

3. Transduction

Zinder and Lederberg (1952) discovered Transduction in *Salmonella typhimurum*. Phage mediated DNA transfer is called Transduction (Figure 1.14).

Transduction is of two types

(i) Generalized transduction (ii) Specialized or Restricted transduction

(i) Generalized Transduction

The ability of a bacteriophage to carry genetic material of any region of bacterial DNA is called generalised transduction.

(ii) **Specialized or Restricted Transduction**

The ability of the bacteriophage to carry only a specific region of the bacterial DNA is called specialized or restricted transduction.

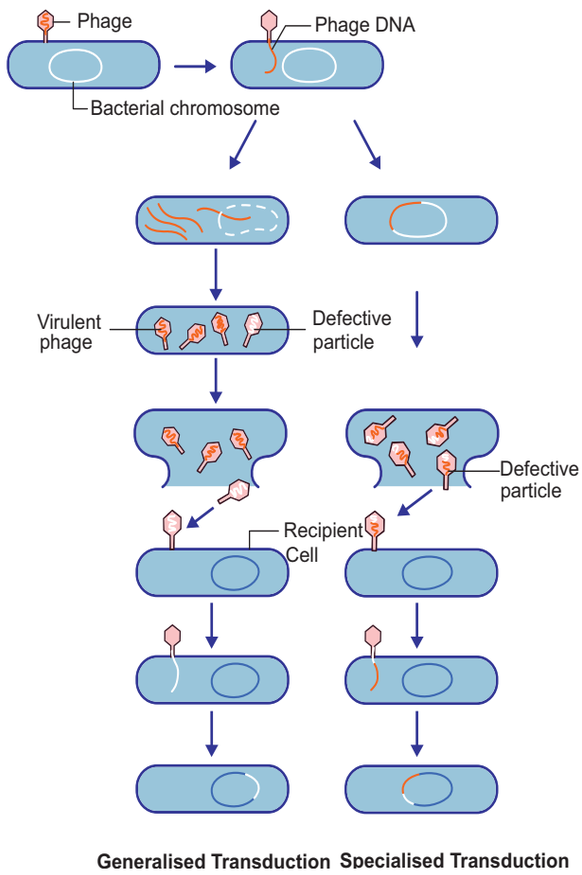


Figure 1.14: Transduction in Bacteria

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